□ 1. Network Addressing – IPv4 and IPv6

□ IPv4 (Internet Protocol version 4)

- **32-bit** addressing □ provides around **4.3 billion** unique addresses.
- Written in **dotted-decimal notation** (e.g., 192.168.0.1)
- **Classes**:
  - Class A: 1.0.0.0 to 126.255.255.255
  - Class B: 128.0.0.0 to 191.255.255.255
  - Class C: 192.0.0.0 to 223.255.255.255
  - Class D (Multicast): 224.0.0.0 to 239.255.255.255

- Class E (Experimental): 240.0.0.0 to 255.255.255.255

- **Private IP Ranges**:
  - Class A: 10.0.0.0/8
  - Class B: 172.16.0.0/12
  - Class C: 192.168.0.0/16

- **Subnetting**:
  - Divides large networks into smaller networks using subnet mask (e.g., /24)
  -

255.255.255.0 ▯ 256 addresses per subnet

▯ IPv6 (Internet Protocol version 6)

- **128-bit** addressing ▯ approx **3.4×10^{3}▯ addresses**
- Written in **hexadecimal** and colon-separated (e.g., 2001:0db8:85a3::8a2e:0370:7334)
- Designed to solve IPv4 exhaustion
- **No need for NAT**
- Types of IPv6 addresses:
    - **Unicast**: One-to-one
    -

Multicast: One-to-many

- **Anycast**: One-to-nearest (routing-based)

- **Loopback Address**: ::1
- **Link-local Address**: fe80::/10 (used for local communication between devices)

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2. Using OpenSSH for Network Communications

OpenSSH is a suite of secure networking utilities based on the SSH protocol.

Key Components:

- ssh: Secure remote login

- scp: Secure file copy
- sftp: Secure FTP
- sshd: SSH Daemon (server)

🔐 SSH Default Port:

- TCP **port 22**

⚙️ Configuration:

- Server config: /etc/ssh/sshd_config
- Client config: ~/.ssh/config

🔐 Authentication:

- **Password-based**
- **Public/Private key pairs** (RSA/ED25519)

  bash

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  ssh-keygen -t rsa ssh-copy-id user@host

🔧 Common Usage:

bash

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```
ssh user@remote-ip scp file.txt user@host:/path sftp user@remote-ip
```

□ 3. Using VNC for Network Communication

VNC (Virtual Network Computing) is used for remote **graphical desktop sharing**.

□ Key Components:

- **VNC Server**: Runs on remote machine
- **VNC Viewer/Client**: Used to access the server
- Default port: **TCP 5900 + display number** (e.g., :1 □ port 5901)

□ Popular VNC Software:

- **TigerVNC**
- **RealVNC**
- **TightVNC**

🔹 VNC on Linux:

- Start server:

  bash

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  ```
  vncserver :1
  ```

- Stop server:

  bash

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  vncserver -kill :1

🔒 Security:

- VNC is **not encrypted** by default.
- Use **SSH tunneling**:

  bash

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```
ssh -L 5901:localhost:5901 user@remote
```

□ 4. Network Authentication

Network authentication ensures that users/devices accessing a network are **authorized**.

□ Types of Authentication:

- **Password-based**
- **Certificate-based**
- **Two-Factor Authentication (2FA)**

🔧 Tools/Protocols:

1. **LDAP (Lightweight Directory Access Protocol)**
   - Directory-based authentication service
   - Used by Active Directory, OpenLDAP
2. **Kerberos**
   - Time-based ticketing protocol
   - Used by Windows domain environments
3. **RADIUS**

- Authentication protocol for remote access servers

4. **SSSD** (System Security Services Daemon)

    - Integrates with LDAP, Kerberos
    - Configured in /etc/sssd/sssd.conf

PAM (Pluggable Authentication Modules):

- Used to integrate local Linux authentication with external services
- Configurable in /etc/pam.d/